



Effects of Spanloading and Slew Angle on an Oblique Flying Wing

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Designing tail-less aircraft has historically involved vehicle stability challenges, leading to issues further in the design process. Motivated from the work of the Horten brothers and Prandtl, this research aims to take an initial step at exploring an Oblique Flying Wing configuration (OFW) with different geometric twist distributions to understand the impact on the stability of the moment coefficients and aerodynamic performance. Using methods from Remiar Horten, parametric variations in spanwise geometric twist were used to create bell-shaped lift distribution in an attempt to produce proverse yaw. A NASA Euler flow solver, FUN3D (Fully Unstructured Navier-Stokes), was used to isolate the aerodynamic components, specifically the yawing moment coefficient, to characterize the impact of slew angle and twist distribution on each aerodynamic component. The aerodynamic center on the asymmetric wing was determined for the baseline and twisted configurations at different slew angles using a Basin-hopping optimization routine with a Sequential Least Squares Programming (SLSQP) subroutine. The results indicate the inclusion of twist reduces the magnitude of the moment coefficient variation with respect to C_L which improves lateral and directional stability, but there are some compromises in aerodynamic performance. While increasing aerodynamic performance is important, the reduction in the moment coefficients may prove to be more advantageous over a given flight envelope. Ultimately, analysis of the twisted configuration was not able to match the bell-shaped lift distribution and confirm the production of proverse yaw, however, the twisted configuration did show the moment coefficients can be manipulated significantly by a twist distribution, improving the stability of the OFW configuration.

I. Nomenclature

C_L	=	Lift Coefficient	$C'_L \cdot c / C_L$	=	Normalized local Lift Coefficient
C_D	=	Drag Coefficient	$C'_D \cdot c / C_L$	=	Normalized local Drag Coefficient
C_Y	=	Side Force Coefficient	c	=	Chord (m)
C_l	=	Rolling Moment Coefficient	$\bar{c}$	=	Mean Aerodynamic Chord (m)
C_m	=	Pitching Moment Coefficient	b	=	Span (m)
C_n	=	Yawing Moment Coefficient	Λ	=	Slew Angle (°)
C_l / C_L	=	Normalized Rolling Moment Coefficient	t / c	=	Thickness to chord ratio
C_m / C_L	=	Normalized Pitching Moment Coefficient	α	=	Angle of attack (°)
C_n / C_L	=	Normalized Yawing Moment Coefficient	α_i	=	Airfoil incidence angle (°)

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II. Introduction

Considerable thought and effort was given to investigate the implications of oblique flying wings (OFW) for supersonic commercial air transport for most part of the 20th century by NASA. These investigations, primarily led by R.T. Jones [1], found that the OFW offered increased aerodynamic efficiency (L/D) when compared to a traditional swept-back wing design in transonic flight conditions. Jones [2] also found that for a given planform area and sweep/slew angle, the oblique wing design yields lower wave drag in the transonic and supersonic regimes for the wing alone, not accounting for any interactions with the fuselage or other components.

However, a major concern of the OFW configuration was found to be stability and control. With the absence of conventional tails, elevators, and rudder control effectors, new methods were introduced to achieve required stability and control of the OFW. To achieve stability, Van Der Valden [3] showed the necessity to use the wing planform for control of a supersonic transport OFW. In order to limit the pitching moment of the passenger OFW, as there are no horizontal tail control surfaces, dihedral was added to the planform to limit the loading on the trailing edge by maintaining consistent lifting pressure along the chord [2, 3]. Additionally, the weight of the OFW must be minimized due to the increased pitching moment caused by the increased lift requirement for trimmed flight. Even with multiple control surfaces, the OFW will not be able to reject upward gusts if the loading on the wing is too high.

Van Der Valden also found that trim in yaw and engine-out control would be a challenge with the lack of a vertical tail. Some solutions for the engine-out control were to have four pivoting engine nacelles to aid in engine-out control along with vertical surfaces attached to the engine nacelles in order to increase the yawing moment from pivoting the engine nacelles [3, 4]. Yaw trim can be achieved with thrust vectoring by pivoting the nacelles and differential thrust, which minimizes penalty due to drag. Others have proposed the use of clamshell ailerons to act as rudder like control surfaces, but were found to lose effectiveness at higher slew angles and will greatly contribute to drag [5]. An alternative method to vertical surfaces on engine nacelles or clamshell ailerons is to tailor the lift distribution to control the yawing moment by changing the spanwise lift distribution. If the spanwise lift distribution is appropriately manipulated, upwash will be produced at the wingtips, generating induced thrust, which can be controlled with conventional ailerons. This allows for yaw control with minimal drag penalty as aileron deflections will help to contribute to, if sized and placed properly, induced thrust rather than induced drag [6].

The effects of upwash at the wingtips was explored by Prandtl as early as 1933 where he theorized, for a given structural weight, a bell shaped lift distribution yields the minimum induced drag solution [7]. Compared to Prandtl's previous work on minimum drag solutions for elliptic planforms, the tapered planform also considers a constraint of minimal wing root bending moment. Prandtl suggested that the tapered planform is most beneficial for an aircraft which does not have any span restrictions, as the tapered wing will increase the span of the designed wing. For the most part, Prandtl's second lift distribution failed to gain much traction with aircraft designers in part due to the complexity required to construct a tapered wing with washout.

During the time Prandtl was developing his second lift distribution, German sailplane designers, Walter and Reimar Horten, were arriving at a similar result with their tail-less flying wing designs. The Horten brothers discovered that a washout twist distribution with taper would yield a stable airplane [8]. The washout twist distribution tailored the lift distribution so that in a turn, the wingtips would produce proverse yaw, or induced thrust, rather than adverse yaw, eliminating the need for a vertical surface. In 1935, the Horten brothers used the proverse yaw in order to achieve positive lateral and directional stability for their flying wing configurations. Their first airplane without any vertical surfaces was the H.II, which had a highly tapered, swept, and twisted planform which likely generated proverse yaw at the wingtips. This allowed for coordinated turns to be achieved without rudder, showing a more efficient way to control a tail-less airplane. While the Horten brothers were the first to coin the 'bell-shaped' lift distribution term, they had a limited understanding on how proverse yaw was generated and were never able to prove its existence [6, 8]. Nevertheless, they built many different flying wing configurations with the bell-shaped lift distribution achieving countless stable flights.

After the Horten brothers, there were few designers that employed the bell-shaped lift distribution. Since the concept of proverse yaw was not well understood, it was difficult to achieve a bell shaped lift distribution without a complex planform. Bowers et. al [6] readdressed the Horten H.II flying wing in 2016 where they showed how proverse yaw can be used for directional control. Their research indicated that the ailerons could be designed to produce varying proverse yaw for coordinated turns without the need of a vertical control surface. It also gave insight into the formation flight of soaring birds which is highly dependent on proverse yaw and spanload manipulation [6]. From the work of Bowers, the underlying mechanism of proverse yaw is better understood, but has primarily been investigated on symmetric, tail-less flying wings. While this configuration has been proven countless times by the Horten brothers and then Bowers, it has not been applied to an asymmetric configuration, or an oblique flying wing, which is the target configuration for this

study.

For this study, the OFW configuration was chosen as the baseline planform to explore due to the historical aerodynamic performance and the difficulty in developing a stable planform. Initially, two designs are investigated, a baseline planform and a twisted configuration developed using methods based on the work of the Horten brothers. Unlike the Horten brothers and Bowers, the planform was not changed to incorporate different taper and sweep design parameter changes. In future work, the planform will be allowed to change in order to achieve a more optimal design, but for this study, in order to understand the impact of a twist distribution on the OFW planform, the planform was fixed. Likewise, only a low-subsonic Mach number is considered to date in order to validate the results in the University of Dayton's Low-Speed Wind Tunnel in the future. With validation, the flight condition space can be further explored to fully understand the aerodynamic performance and impact of the twist distribution, which will buy-down the risk of any follow on high-speed testing.

III. Methodology

A. Model generation

Engineering SketchPad (ESP), an open source parameterized geometric modeler, is used as the geometry generator as it allows for rapid generation of various computational models with associative geometry [9]. The planform of the OFW, Figure 1, is derived from the DARPA-Nothrop Grumman Switchblade program using similar methods to that found in Plumley [10]. Keeping the planform of the OFW fixed, a parameterization on the geometric twist distribution (local airfoil incidence angle) is created to link with the solved twist distribution. The planform is asymmetric about the center line (defined as the apex of the leading edge), with a leading edge sweep back of 10° . For the full size vehicle, the span is 61 m [11], with left portion of the wing having a semi-span of 34 m and the right portion of the wing having a semi-span of 27 m. The wingtips are highly tapered with the left wingtip having a leading edge sweep of 45° while keeping a straight trailing edge. Conversely, the right wingtip leading edge is swept back 10° , the same as the rest of the wing, but the trailing edge of the right wingtip is swept forward 45° . The root chord of the OFW (apex location) is 12.4 m, and using the root chord to define the leading and trailing edge locations, the spanwise chord distribution is determined.

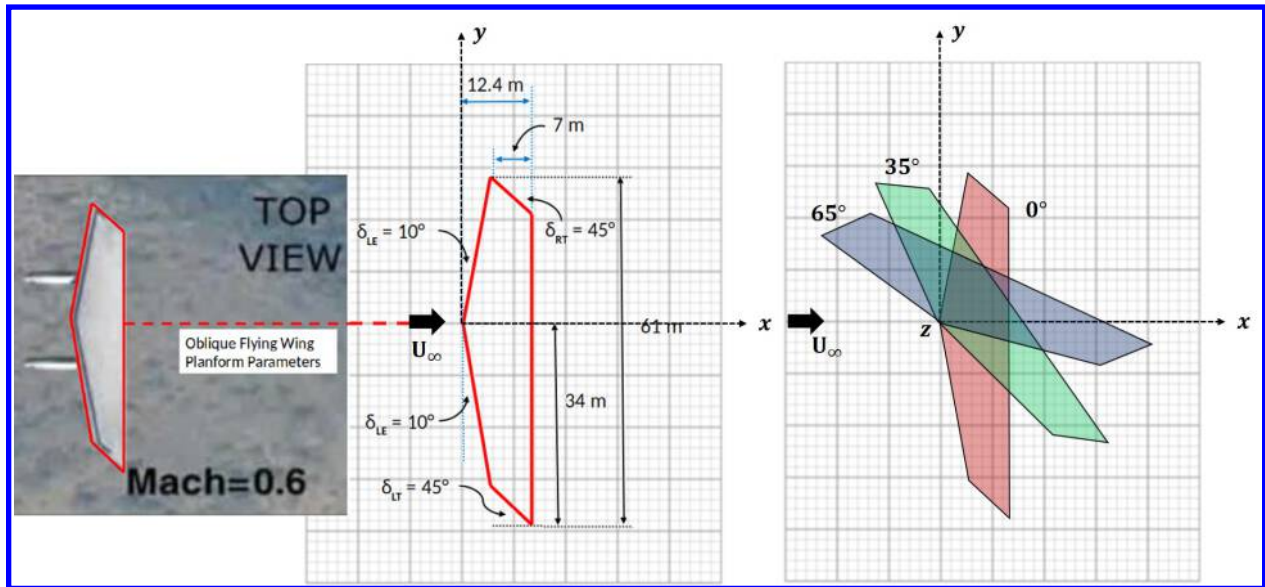


Fig. 1 Planform view of artistic representation of Northrop Grumman/DARPA Switchblade OFW, digitized, with planform parameters.

With the planform fixed, the various twist distributions along the span can be computed using the method employed by the Horten brothers. Dividing the OFW into 41 spanwise airfoil segments to define the planform geometry, each spanwise airfoil segment contains the following parameters:

- 1) X-Location of Leading Edge
- 2) Y-Location of Leading Edge
- 3) Z-Location of Leading Edge
- 4) Chord length of spanwise section
- 5) Thickness to chord ratio (t/c)
- 6) Incidence angle of airfoil (α_i)

These six parameters are then compiled together in ESP to generate the computational model:

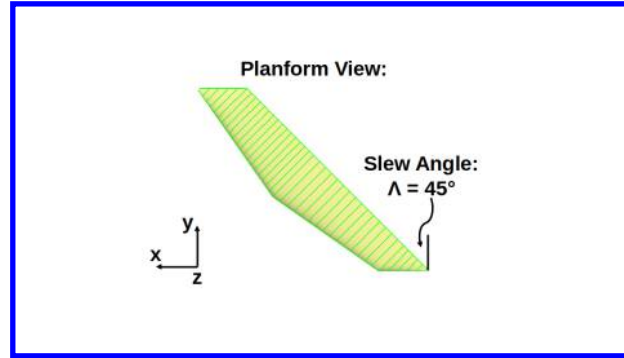


Fig. 2 Engineering Sketch Pad (ESP) model of the OFW with slew and body-axis definitions

When the OFW is slewed, the aerodynamics reference properties, such as the span and mean aerodynamic chord (MAC), change with respect to the slew angle. These changes have to be accounted for when performing the moment coefficient calculations as the changing quantities will give different results based on the configuration. To account for these changes, the span is always defined as:

$$b = y_{max} - y_{min} \quad (1)$$

where the y_{max} and y_{min} values are calculated by ESP during the model generation. For the MAC, methods described by Vogeltanz uses a series of trapezoids to determine the MAC and its leading edge x,y location on the wing [12]. To facilitate the calculation, the OFW's planform was manually divided into 4 trapezoids for each slew angle after the models were generated. Then using Vogeltanz's method, the leading edge x,y location of the MAC was determined and is shown in Figures 3a and 3b.

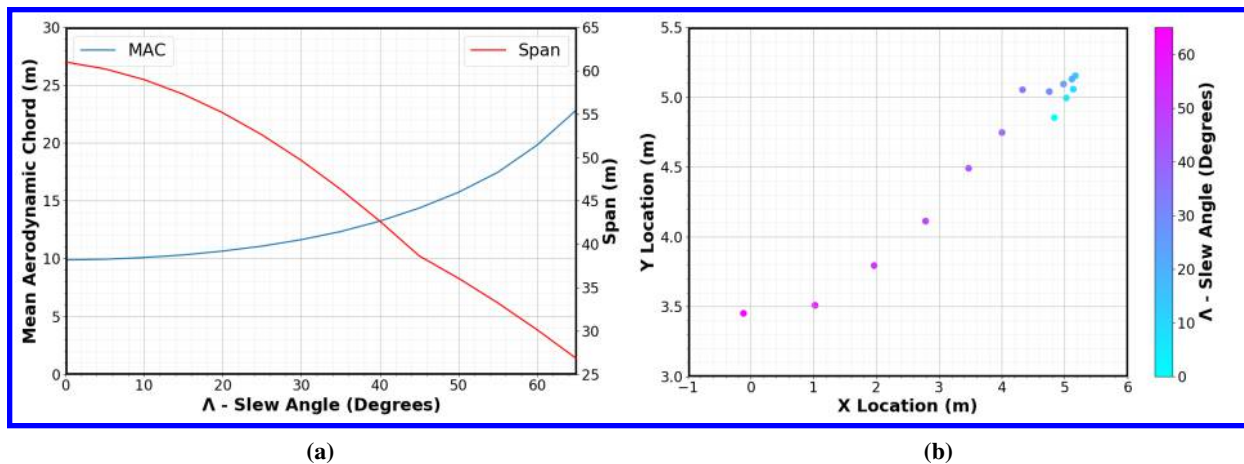


Fig. 3 a) Change in mean aerodynamic chord and span with increasing slew angle b) X and Y location of the leading edge of the mean aerodynamic chord

As expected, the span decreases and the MAC increases with increasing slew angle. The MAC also changes as a function of the slew angle, however, at lower slew angles, the change in MAC is comparatively less. It is not until a slew angle of $\Lambda \approx 35^\circ$ until the MAC leading edge location begin to significantly change, which is caused by more of the lifting surface area moving to the right due to the increasing slew angle, effectively balancing the lifting area about that point.

The MAC leading edge x and y reference locations are used as the initial moment reference location for each slew case along with the calculated MAC and span from each model. It is assumed that the MAC and span are independent of twist distribution as the 2-D planform does not change with varying twist distributions. The Clark-Y airfoil (Figure 4) with a constant thickness to chord ratio (t/c) of 12.0% was chosen to make up the entire span of the OFW as it has positive camber which is beneficial when defining the twist distribution in order to meet the sectional lift requirements while limiting the angle of attack.

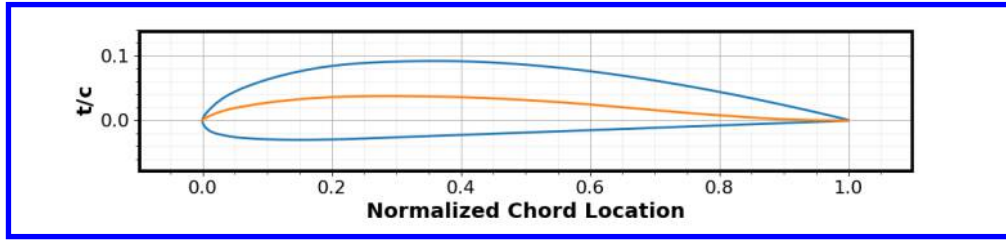


Fig. 4 Clark-Y Airfoil and the mean camberline

B. Geometric Twist Distribution Determination

Reimar Horten's method involves determining the 'bell-shaped' lift distribution using basic lifting line theory. Using a predetermined design lift coefficient of $C_L = 1.0$, the corresponding geometric twist distribution can be calculated. Using the parametric equation for an ellipse:

$$\cos(x) + \sin(x) = 1 \quad (2)$$

Equation (2) will result in an elliptical distribution, but Horten uses $\sin^n(x)$ for his distribution, where n is varied to obtain a bell-shaped lift distribution seen in Figure 5a.

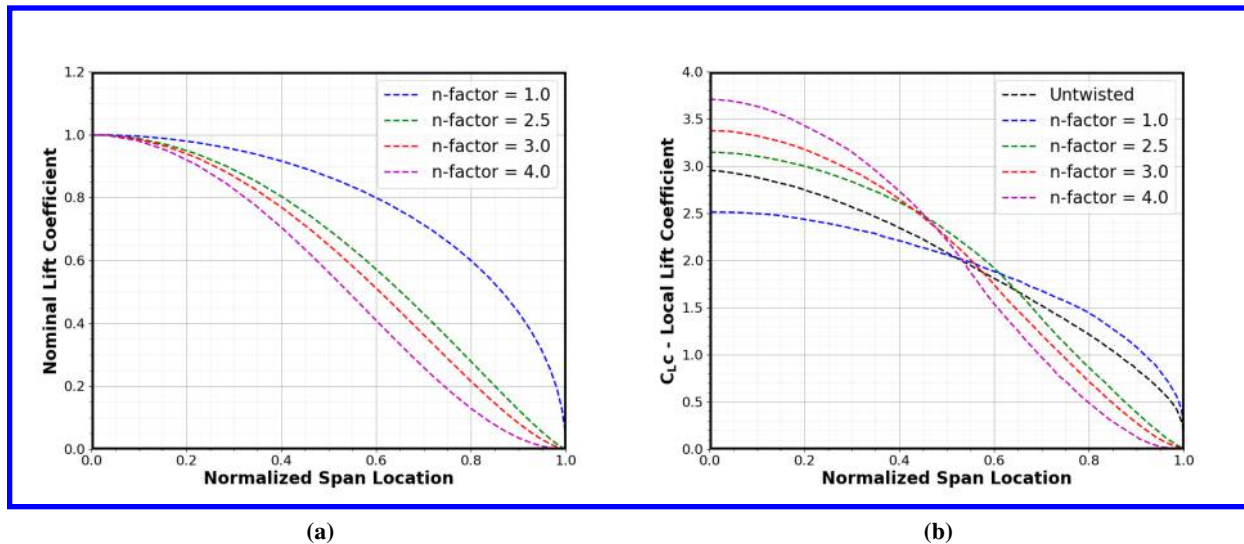


Fig. 5 a) Arbitrary semi-span lift distributions created using parametric equation for an ellipse with exponential factors b) Local lift coefficient (C_{Lc}) for the H.II flying wing. Shows various spanloading distributions derived from Horten's method [8]

Using Horten's rendition of the parametric equation, the local lift coefficient can be solved for using the following equation:

$$C_L \cdot c(x) = \int cte \cdot \sin^n(x) dx \quad (3)$$

Where cte is an influence coefficient and x is the normalized spanwise station of the airfoil section. From Horten, the various local lift distributions for a half-span of the H.II airplane can be seen for notional n factors in Figure 5b. Equation (3) can then be used to find the designed lift coefficient for a given angle of attack and the influence coefficient, cte , by dividing through Equation (3) by the chord of each spanwise station. The influence coefficient is iteratively changed so that the lift coefficient solution matches the design lift coefficient.

$$C_L = \int \frac{cte \cdot \sin^n(x)}{c(x)} dx \quad (4)$$

Knowing the influence coefficient allows for the determination of the local lift coefficient for each airfoil section along with the corresponding incidence angle of each airfoil section. Using 2-D linear lift coefficient properties of the Clark-Y airfoil at a Reynolds number of 250,000, the incidence angle, α_i , of each airfoil section was determined using the following method:

$$C_{Lc} = \frac{d\alpha}{dC_L} \alpha_i + C_{La_0} \quad (5)$$

Where C_{Lc} is the local lift coefficient of the airfoil section, $\frac{d\alpha}{dC_L}$ is the lift curve slope of the desired airfoil, and C_{La_0} is the lift coefficient at a 0° angle of attack for the desired airfoil. Using a linear fit between the airfoil sections, the twist distribution can be defined for a given planform where the span location, chord length, and airfoil are known. For the OFW, n factors of 1, 2, 2.5, 3, and 4 were chosen in order to understand how the differences in lift distribution affect the twist distribution of the OFW. The resultant local lift coefficient distributions for each n factor are shown in Figure 6a.

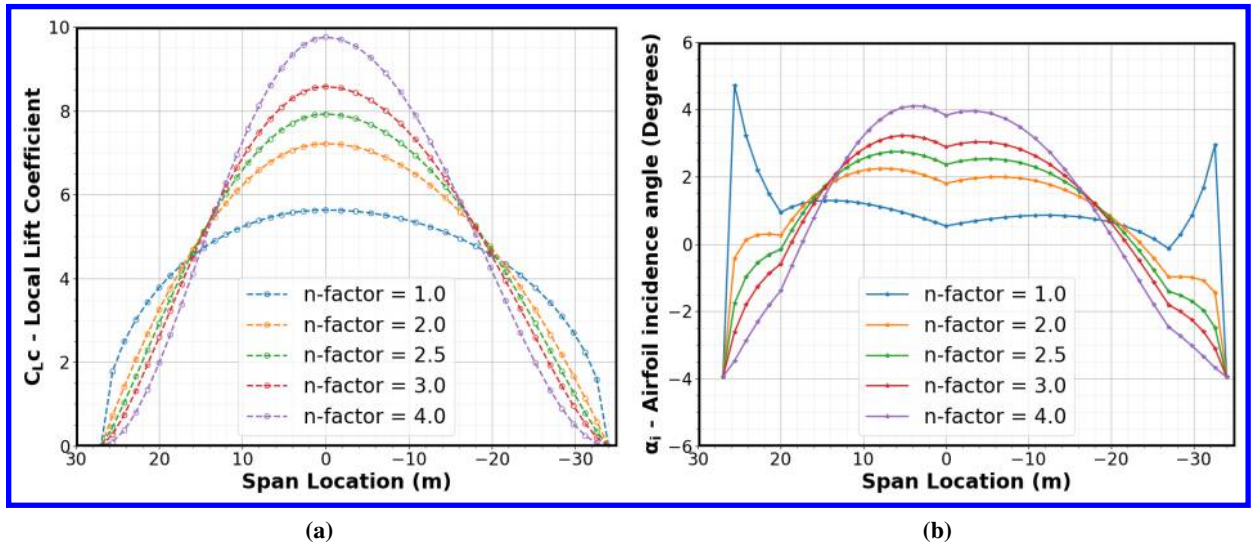


Fig. 6 a) The local lift (C_{Lc}) distribution for the OFW with different n factors with a design lift coefficient of $C_L = 1.0$. b) The resultant geometric twist distribution profile for the OFW.

Figure 6b shows the resulting incidence angle for each airfoil section according to Equation (5). Moving from root to tip, the incidence angle varies significantly, especially at the tip where the incidence angle goes to: $\alpha_i \approx -4^\circ$. For the current study, only the $n = 2.5$ twist distribution and baseline untwisted case are chosen for analysis. The Horten brothers identified that the $n = 2.5$ twist distribution brought about the ideal lift distribution profile for yaw stability.

IV. Computational Setup

A. Computational Analysis

To understand the sensitivity of the moment coefficients, FUN3D (Fully Unstructured Navier-Stokes) with the incompressible Euler equations is used to solve for the forces and moments on the OFW [13]. FUN3D, for the inviscid equations, requires a reference temperature, which was set to $T = 293$ K, and a Mach number of $M = 0.1$ was used to allow for comparison with subsonic wind tunnel testing in the future. The OFW and fluid-domain is modeled in ESP, which through CAPS (Computational Aircraft Prototype Synthesis), takes the relevant geometric information to pass to the automated meshers, AFLR (Advancing-Front/Local Reconnection Unstructured grid generator) 3 and 4 [14–16].

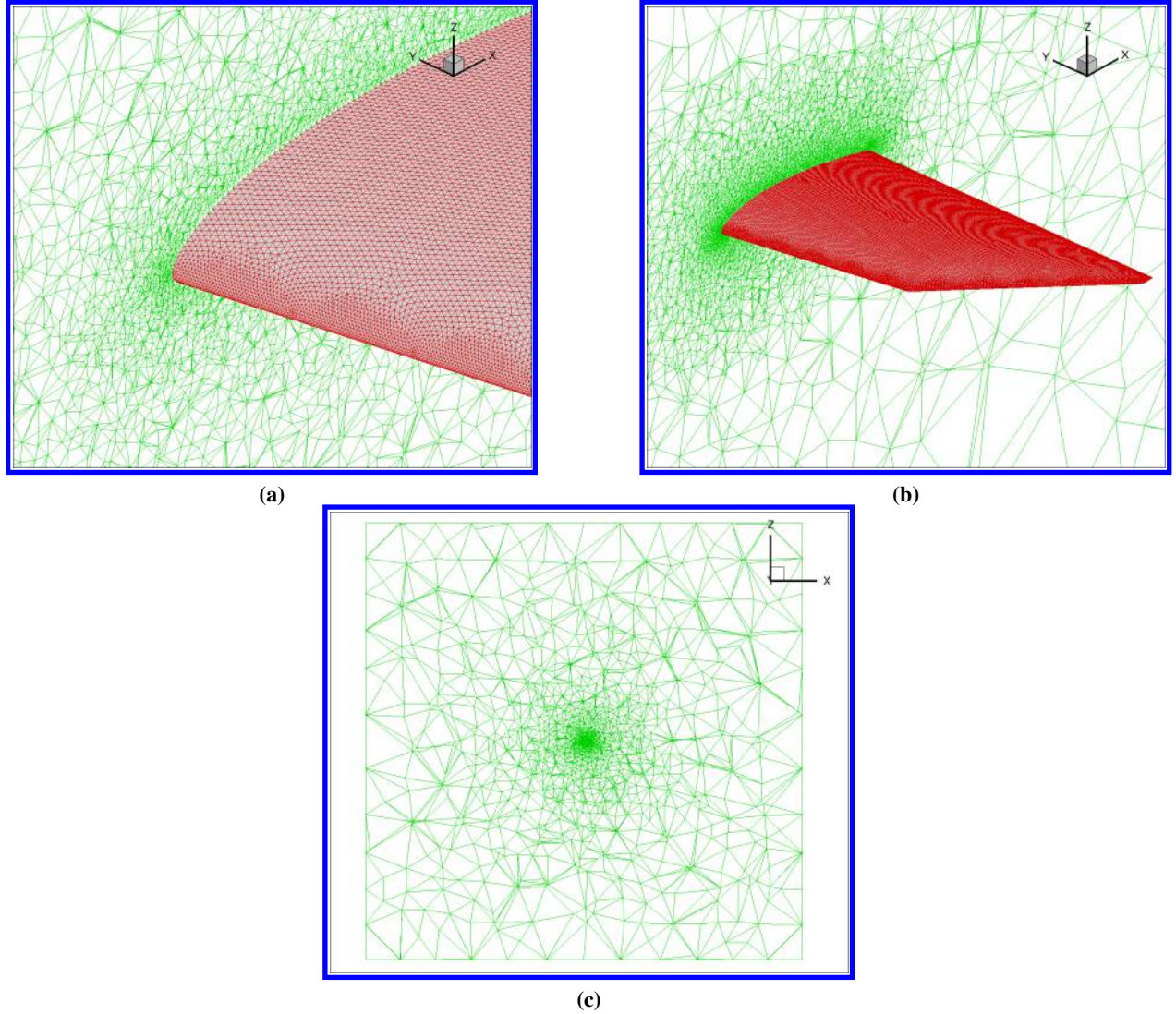


Fig. 7 Medium grid: a) Surface and volume mesh slice showing meshing resolution of leading edge b) Surface and volume mesh slice showing portion of the left wing c) Farfield mesh showing grid refinement towards the surface

The meshing parameters are attributes of the geometry, with finer mesh control being housed in CAPS. Figures 7a and 7b show the medium grid that was generated using the automated meshing parameters. A mesh resolution study is performed by varying the meshing parameters in CAPS to generate different mesh refinements, with the mesh refinement being driven by the surface meshing parameters. Solutions for the different meshes were run until convergence (residuals reducing by six orders) to determine what mesh density is needed for the best computation time and solution accuracy.

An initial coarse mesh of 270,000 nodes was used as a baseline, and then using the meshing parameters in CAPS, the surface mesh was globally refined to reach a desired node count. Table 1 shows the number of nodes for each level of mesh that was generated.

Table 1 Mesh refinement study

Mesh	Number of nodes
Coarse Mesh	270,000 nodes
Medium Mesh	975,000 nodes
Fine Mesh	2,690,000 nodes

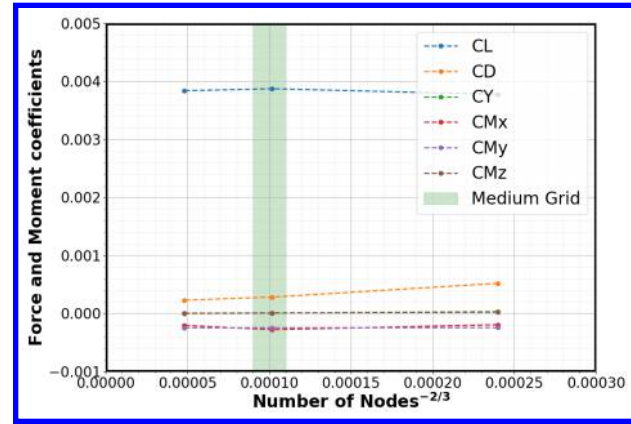


Fig. 8 Mesh convergence study using AFLR3/4 automated meshing

Taking the untwisted, unslewed OFW case as a baseline, results from each generated mesh were compared to analyze the computational performance of the meshes and the trends in the forces and moments. The resulting mesh convergence study shows that with mesh refinement, the solution changes very little from the medium mesh to the fine mesh, indicating that the solution would not change appreciably with further mesh refinement. The parameters from the medium mesh (975,000 nodes) were chosen for use throughout the computational study as the medium mesh demonstrates reasonable convergence in the solution, with significantly less computational costs. With the medium mesh, a total of 252 solutions are needed to explore the effects of slew angle and angle of attack for the baseline and twisted OFW models. Given the significant number of solutions needed in order to fully explore the design space, decreased computational time and cost is desired. Table 2 shows the test parameters and reference quantities that were used for the computational study in FUN3D.

Table 2 CFD analysis matrix with configuration parameters and moment reference locations

Slew angle (Λ)	Angle of attack (α)	MAC - $\bar{c}$ (m)	Span - b (m)	x (m)	y (m)	z (m)
0°	-4° to 12°	9.88	61.00	4.84	4.85	0.00
5°	-4° to 12°	9.95	60.21	5.03	4.99	0.00
10°	-4° to 12°	10.10	58.97	5.14	5.06	0.00
15°	-4° to 12°	10.32	57.28	5.17	5.15	0.00
20°	-4° to 12°	10.65	55.15	5.12	5.13	0.00
25°	-4° to 12°	11.08	52.60	4.98	5.09	0.00
30°	-4° to 12°	11.64	49.65	4.76	5.04	0.00
35°	-4° to 12°	12.34	46.32	4.33	5.05	0.00
40°	-4° to 12°	13.25	42.64	4.00	4.75	0.00
45°	-4° to 12°	14.37	38.64	3.47	4.49	0.00
50°	-4° to 12°	15.74	36.06	2.79	4.11	0.00
55°	-4° to 12°	17.47	33.22	1.97	3.79	0.00
60°	-4° to 12°	19.83	30.12	1.02	3.51	0.00
65°	-4° to 12°	22.87	26.79	-0.11	3.45	0.00

V. Results

The primary objective of this body of research is to further understand the relationship between the slew angle of an OFW and geometric twist distribution on the yaw moment coefficient. Results from inviscid, incompressible CFD are used to isolate each component of interest, primarily the aerodynamic performance and the change in the moment coefficients with slew angle and twist distribution. Lift and drag coefficients along the span are used from the FUN3D analysis to better understand the role of the twist distribution on the total vehicle force and moment coefficients, along with the local forces.

A. Force Coefficient Results

The tested OFW models consisted of two different geometric twist distributions, one with no geometric twist, and another with a twist distribution which was determined using methods developed by the Horten brothers. These different twist distributions will inherently lead to different spanwise lift coefficient profiles, but should give similar lift curve slopes as they have the same airfoil. Figure 9a shows that while the different twist distributions lead to different zero-lift angles of attack, their respective lift curve slopes are nearly identical. For increasing slew angle, the difference in the zero-lift angle of attack decreases as the characteristic chord length of the airfoil increases.

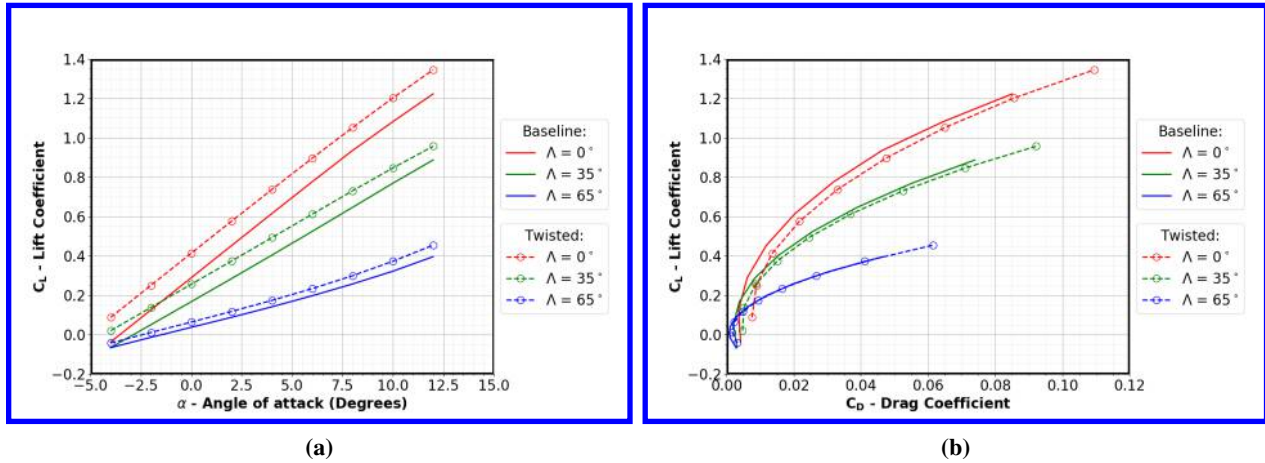


Fig. 9 a) Lift coefficient for three slew angles comparing the baseline and twisted OFW configuration b) Lift and Drag (induced and pressure) coefficient polar for three slew angles comparing the baseline and twisted OFW configuration

The decreasing lift curve with slew angle is a function of decreasing aspect ratio, decreasing span, and the change in the airfoil orientation with respect to the freestream. The decreasing aspect ratio and decreasing span affect the lift curve the most compared to the change in airfoil orientation. A caveat of the analysis performed on the OFW configurations is that it does not account for any stall effects. It is unlikely that the lift curve slope will remain constant at higher angles of attack, especially for the twisted OFW, which will surely be stalled at the root for higher angles of attack given the airfoil incidence angle at the root.

The drag polar in Figure 9b shows the same trend as seen in Figure 9a. For the slew angle of $\Lambda = 65^\circ$, the drag polar is nearly identical, with deviation occurring at lower slew angles. This trend can also be seen for lower slew angles as well. Like the lift coefficient, viscous effects are neglected, changing the higher angle of attack performance. Including the viscous effects, there is likely to be a change in the drag coefficient, especially when the twisted OFW configuration begins to stall as it will greatly increase the drag on the OFW.

Increasing slew angle shows that the differences in both the lift coefficient and drag polar decrease, with the $\Lambda = 65^\circ$ slew case having almost identical performance for the drag polar. These changes in the performance differences can be explained by the differences in the thickness to chord ratio caused by the changes in airfoil orientation. Typically, conventional airplane configurations align airfoil sections to be streamwise, or parallel with the flow. There are some exceptions to this rule in the case of swing-wing airplanes or airplanes with oblique wings like the NASA F-8 Oblique Wing Research Airplane and the OFW. As the wing is slewed, the streamwise flow no longer sees the predetermined airfoil shape, leading to a change in the aerodynamic performance for the thinner, reoriented airfoils. Figures 10a and 10b show the airfoil profiles and how the characteristic chord length of the airfoil increases at the $x = 0$ m location.

There are clear differences in the airfoil incidence angle for the $\Lambda = 0^\circ$ slew case in Figure 10a, but at the $\Lambda = 65^\circ$ slew case, Figure 10b, the differences are not in the airfoil incidence angle, but rather the airfoil thickness and camber distribution. These changes in the airfoil shape that the flow will see impact the performance differences between the baseline and twisted OFW configurations, but at the moment, slewing the wing appears to reduce the changes in incidence angle, which could prove to be beneficial at higher Mach numbers.

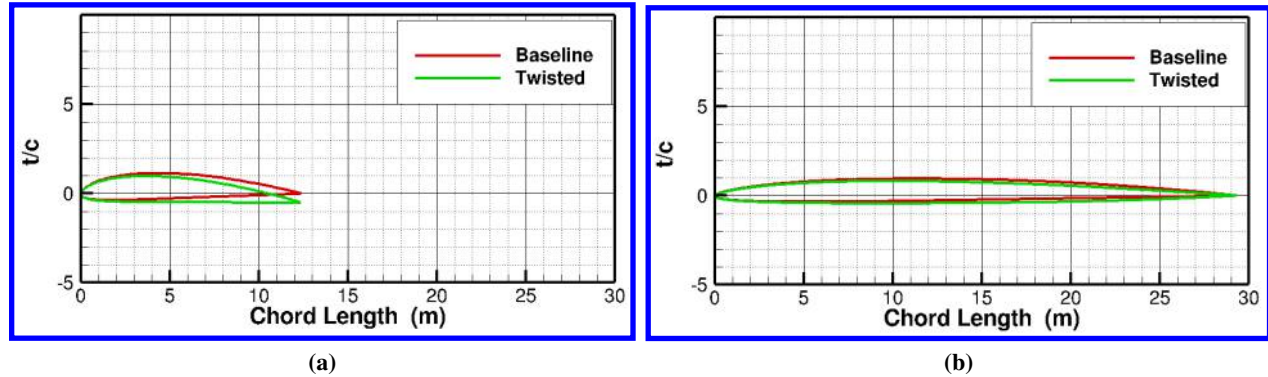


Fig. 10 Airfoil profiles taken at $y = 0$ (m) for the baseline and twisted OFW at a slew angle of: a) $\Lambda = 0^\circ$ b) $\Lambda = 65^\circ$

The maximum aerodynamic efficiency, L/D , changes with slew angle, with a slew angle of $\Lambda = 0^\circ$ showing the greatest aerodynamic efficiency, as this slew angle has the highest aspect ratio, contributing to the aerodynamic efficiency. As the slew angle is increased, the max aerodynamic efficiency decreases for the baseline configuration, but stays relatively constant for the twisted configuration. Comparing the twisted and baseline OFW configuration, the twisted OFW shows a reduction in the untrimmed aerodynamic efficiency and then converges back towards the baseline's aerodynamic efficiency. At an angle of attack of $\alpha = 0^\circ$, the difference in maximum L/D between the baseline and twisted configuration is 27.9%. For $\alpha = 8^\circ$, the difference is less apparent between the configurations, with the difference being 14.8%.

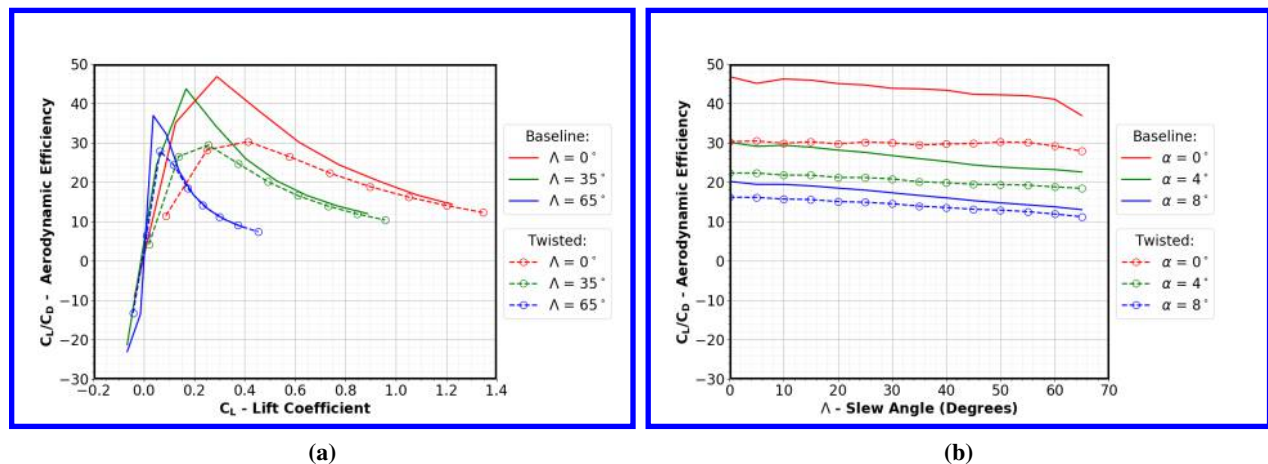


Fig. 11 Aerodynamic efficiency comparing the baseline and twisted OFW configuration: a) For three slew angles b) As a function of slew angle

While the aerodynamic efficiency is an important consideration, the benefits of the twisted configuration on the moment coefficients may outweigh the decrement in aerodynamic efficiency. If other factors are considered such as stability, control, and trim, the twisted configuration may benefit by decreasing the impact of the moments for a level flight condition, likely reducing the trim drag for that condition and potential increasing the controllability of the vehicle. At this point in the study, the extent of the difference in aerodynamic efficiency with respect to the moment coefficients is not yet determined.

B. Aerodynamic Center Results

Understanding the sensitivities of the moment coefficients with changing twist distribution and slew angle is critical when it comes to developing a stable platform for the OFW. The aerodynamic center plays a large role in understanding the moment coefficients with different slew angles and twist distributions. Initially, the x , y , and z reference locations for FUN3D analysis were determined based on the location of the MAC leading edge found using Vogeltanz's method. This acts as an initial position for the aerodynamic center, but the true aerodynamic center needs to be determined post-analysis. For the OFW, the aerodynamic center is defined as the location where all of the moment coefficients do not change significantly with a change in angle of attack:

$$\frac{\partial C_{l_{ac}}}{\partial \alpha} \approx \frac{\partial C_{m_{ac}}}{\partial \alpha} \approx \frac{\partial C_{n_{ac}}}{\partial \alpha} \approx 0 \quad (6)$$

Using a rigid body moment reference translation matrix [17]:

$$\begin{bmatrix} C_{l_{ac}} \\ C_{m_{ac}} \\ C_{n_{ac}} \end{bmatrix} = \begin{bmatrix} C_l \\ C_m \\ C_n \end{bmatrix} - \begin{bmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{bmatrix} \cdot \begin{bmatrix} -C_D \\ C_Y \\ -C_L \end{bmatrix} \quad (7)$$

The x , y , and z moment reference location can be systematically changed to find the aerodynamic center based on the force and moment coefficients from FUN3D. To facilitate the identification of the aerodynamic center, a Scipy implementation of Basin-hopping was used with a Sequential Least Squares Programming (SLSQP) subroutine. For the Jacobian in the SLSQP subroutine, the gradients were approximated using a 2-point finite difference scheme [18]. The moment coefficients were characterized with respect to angle of attack for each slew angle by using a third order polynomial curve fit:

$$f(x) = a_i x^3 + b_i x^2 + c_i x + d_i \quad (8)$$

Where i is with respect to the three moment coefficients. The polynomial coefficients on variable terms, x , are minimized in order to satisfy the definition of the aerodynamic center, Equation (6). If the variable coefficients are minimized sufficiently, they will have a negligible impact on the function, making the constant coefficient d the function value. With this, the aerodynamic center can be found using Equation (9) as it sums the different polynomial functions for each moment coefficient and looks to minimize the coefficients which are dependent on angle of attack.

$$\underset{x,y,z}{\text{minimize}} \quad f(x, y, z) = \sum_i (a_i^2 + b_i^2 + c_i^2) \quad (9)$$

The aerodynamic centers analysis is performed on each configuration as it is expected that the twisted OFW will have a different aerodynamic center compared to the baseline OFW, with location of the MAC leading edge used as the initial guess for both configurations. All optimization cases have a final objective function value of $f(x, y, z) \leq 10^{-6}$, showing a reasonable minimization in the polynomial coefficients, with bounds on the x , y , and z locations so that the aerodynamic center is a physical point on the OFW.

From the aerodynamic center optimization routine, it is clear for at least small slew angles, the MAC leading edge y location is not the aerodynamic center. This is likely caused by the increased moment arm and lift production to the left of the MAC leading edge location, as the optimizer is driving the y location closer to the center of the wing. As the slew angle is increased beyond $\Lambda = 45^\circ$, the MAC leading edge y location is more representative of the y aerodynamic center. For the most part, the baseline configuration shows that the MAC leading edge x location is a reasonable estimation of the x aerodynamic center. In the case of the twisted OFW configuration, the aerodynamic center approximation in the x and y directions are significantly better, especially at lower slew angles. Like the baseline configuration, the x location does not change significantly, with most of the change happening in the y direction. Comparing the two configurations to the Vogeltanz's MAC method, the y location moves on average, 7.76% for the baseline configuration compared to 3.77% for the twisted configuration. The variation in the y location for the aerodynamic center is due to the spanwise loading on each configuration. If the spanwise lift profiles were more similar, then it would be expected that the y aerodynamic center location would be more similar between the two configurations.

The z reference location is important to determine for the aerodynamic center as it plays a large role in the pitch moment, which is important to determine longitudinal stability of the OFW. At low slew angles, less than $\Lambda = 25^\circ$, the z location for the twisted configuration is relatively constant, but is not the case for the baseline configuration. Above $\Lambda \approx 30^\circ$, the z aerodynamic center location trades off between positive and negative, which could be expected as

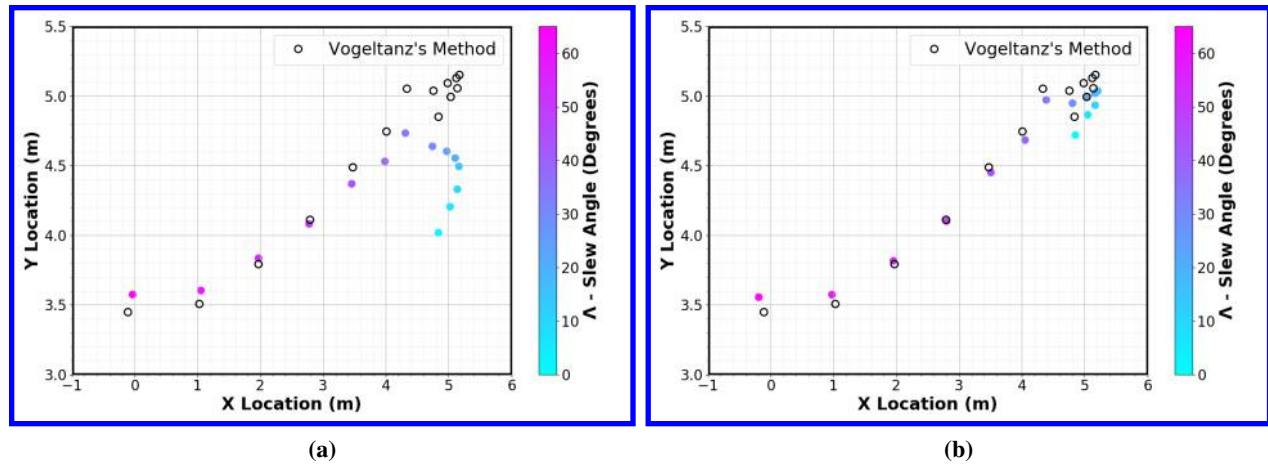


Fig. 12 x and y - Aerodynamic center for a) Baseline configuration comparing initial MAC reference b) Twisted configuration comparing initial MAC reference

the rolling moment coefficient, which is highly sensitive to slew angle, is also dependent on the z reference location. The big take away from examining the z aerodynamic center location is that it was difficult to find a well converged solution without using the z component in Equation 7, showing the difficulty in determining the behavior of the OFW configuration.

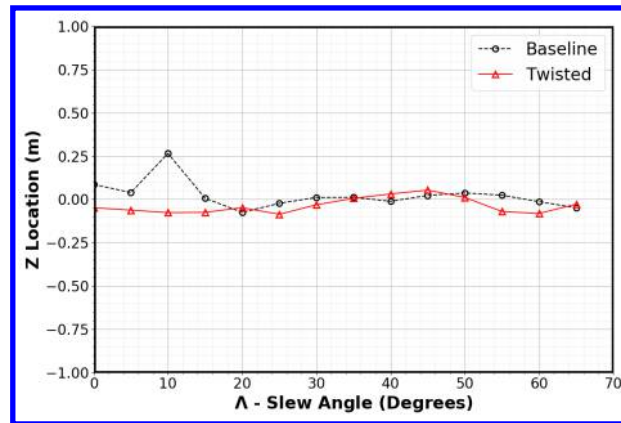


Fig. 13 z - Aerodynamic center for baseline and twisted configuration

C. Moment Coefficient Results

1. Rolling Moment Coefficient

For all of the moment coefficients, the moment reference locations used were the MAC leading edge x and y location with the z location being $z = 0.0$ (m). Table 2 shows the breakdown of the configuration parameters and moment reference locations for the baseline and twisted configurations. Like what was seen in the aerodynamic center identification results, the rolling moment plays a strong role in the OFW configuration, mostly due to the spanwise asymmetries. The asymmetric planform already yields significant challenges as the lift distribution will not be symmetric causing a rolling moment which is unlike traditional configurations. Figure 14a shows that at a $\Lambda = 0^\circ$, there is a positive rolling moment due to the increased lifting planform area to the right of the MAC.

Like the lift coefficient, Figure 14a shows as the slew angle is increased, the differences between the baseline and twisted configuration decrease, and at a slew angle of $\Lambda = 65^\circ$, the differences between the twisted and baseline configuration are indistinguishable. This is likely caused by the increase in the characteristic chord length of the airfoil

that was seen in the lift coefficient as the slew angle increases. Having different airfoil orientations with respect to the freestream will affect the thickness to chord ratio, changing the lift production on either side of the moment reference. These differences are more pronounced at the lower slew angles where the twist distribution is more established and the characteristic chord length of the airfoil has not been changed by slewing the OFW, creating the significant change in the rolling moment. From Figures 10a and 10b, the airfoil shapes become more similar to each other and the aerodynamic center analysis in Figures 12a and 12b shows that the aerodynamic centers are more similar at the higher slew angle cases.

The results from 14a and 14b can be represented from a standpoint of basic flight maneuvers. Due to the high magnitude change in rolling moment with the C_L , the aircraft/wing will experience a significant roll while taking off or landing. To counteract the rolling moment, a large control input would be required, but given the flight condition, this may prove to be challenging. However, if the lift distribution can be altered to what is shown in the Figure 14a, then rolling moment is significantly reduced. This has implications in reducing the trim drag during these maneuvers and indicates that it is aerodynamically possible to take off or land at slewed configurations

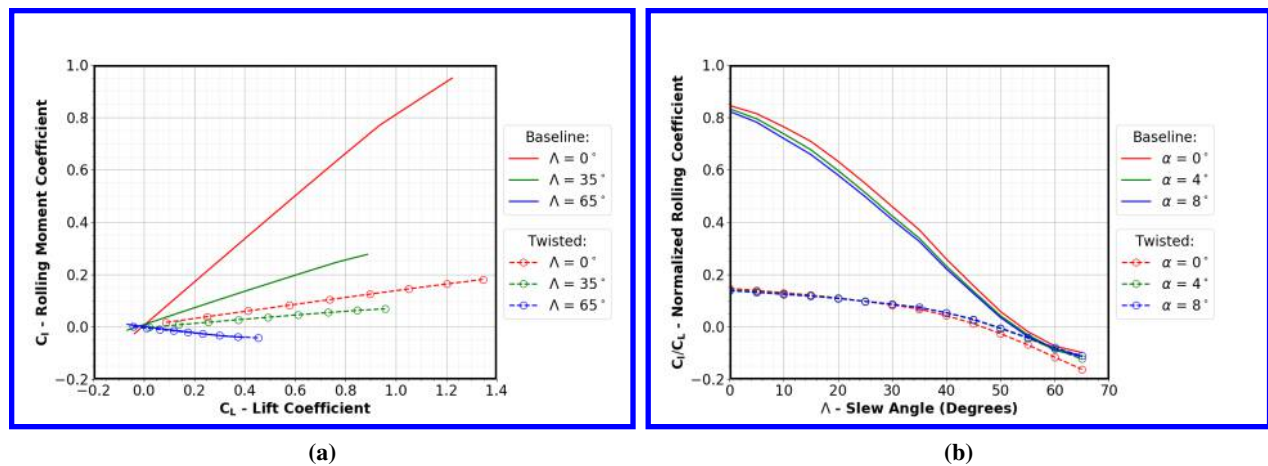


Fig. 14 a) Rolling moment coefficient comparing the baseline and twisted OFW configuration at three slew angles b) Normalized rolling moment coefficient comparing the baseline and twisted OFW configuration at three angles of attack

As the slew angle is increased, the rolling moment decreases, with the decrement being much more apparent in the baseline configuration compared to the twisted configuration. Figure 14b shows that at a slew angle of $\Lambda \approx 50^\circ$, the rolling moment normalized by the lift coefficient, changes sign for the twisted configuration and the same happens at a slew angle of $\Lambda \approx 55^\circ$ for the baseline configuration. The behavior of the rolling moment coefficient with respect to slew angle is very important as it will determine how the OFW configuration can be controlled. An interesting result for both configurations is the insensitivity to slew angle with angle of attack. Figure 14b shows that with increasing slew angle, the normalized rolling moment does not change with angle of attack which allows for the rolling moment to be easily predicted using only angle of attack and slew angle.

The advantage that the twisted configuration shows is that for moderate slew angles, the rolling moment does not change significantly, making the behavior of the OFW more predictable. Additionally, going back to the aerodynamic performance of each configuration, the twisted configuration is starting to show that there is a benefit in taking an efficiency penalty as the rolling moment coefficient magnitude is significantly reduced.

2. Pitching Moment Coefficient

The pitching moment and longitudinal stability will always be a concern for any tail-less air vehicle as controllability is limited due to the absence of a horizontal tail. Typically, an airplane is considered stable in pitch when the pitching moment is able to be trimmed with angle of attack and/or horizontal control surfaces. Like the rolling moment, the baseline and twisted OFW show the same trends in the pitching moment, with discrepancies in the order of magnitude likely caused by the differences in the aerodynamic center location. Unlike the rolling moment coefficient, the twisted configuration shows an increase in the magnitude of the pitching moment compared to the baseline configuration. This change will affect the control surface deflection required in order to trim the pitching moment, in turn increasing the

trim drag which is ideally managed by a horizontal tail with a large moment arm, but this is not the case for the OFW configuration.

For the $\Lambda = 0^\circ$ and $\Lambda = 35^\circ$ slew cases the pitching moment is negative, but for the $\Lambda = 65^\circ$ case, the twisted configuration shows instabilities in pitch at a moderate lift coefficient and then trends negative at higher lift coefficients. While pitch instability is a concern for any tail-less configuration, it is unlikely in a realistic flight envelope that the OFW will fly at a slew angle of $\Lambda = 65^\circ$ and a Mach number of $M = 0.1$. The OFW will need the increased span and aspect ratio to be able to produce the lift required at that Mach number for trimmed flight, making it very unlikely the OFW will fly at a high slew angle for low speed flight. Therefore, to amend what is stated in the previous section, based on pitch stability, it is not possible for the wing to takeoff or land in a highly slewed configuration.

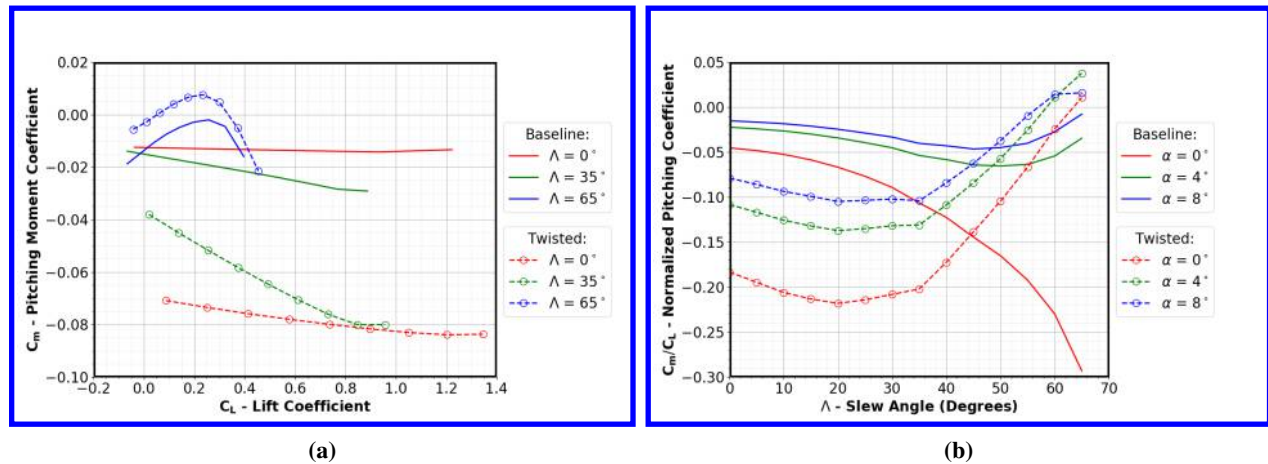


Fig. 15 a) Pitching moment coefficient comparing the baseline and twisted OFW configuration at three slew angles b) Normalized pitching moment coefficient comparing the baseline and twisted OFW configuration at three angles of attack

An interesting difference between the baseline and twisted configuration, Figure 15b, is that with increasing slew angle, trends in the pitching moment are similar for the twisted configuration, but not for the baseline configuration. This is not the case in the baseline configuration which shows that for an angle of attack of $\alpha = 0^\circ$, the trend is opposite respective to the other angle of attack cases, with a greater magnitude. Comparing the sensitivity to slew angle, the twisted configuration shows greater consistency in the normalized pitching moment, while the baseline configuration shows significant changes in the sensitivity of the normalized pitching moment with respect to angle of attack and slew angle. In both cases, simple low order models may not be able to predict the pitching moment due to high variability with angle of attack and slew angle.

3. Yawing Moment Coefficient

The yawing moment coefficient will be one of the primary indicators of proverse yaw generation, as if there is a change in the spanwise distribution of drag, it will be evident in the yawing moment coefficient. Ideally, the change in the yawing moment coefficient will be caused by induced thrust generated at the wingtips, which is the goal of adding in a predetermined twist distribution. From Figure 16a, it is clear that the twisted configuration has a significant impact on the yawing moment coefficient, but ideally, the yawing moment would be $C_n = 0$ for the designed lift coefficient as the twisted configuration should have accounted for the asymmetries in the planform. As this is not the case, it is inconclusive as to whether proverse yaw is being generated by the twisted configuration, but regardless, there are still many benefits that it exhibits compared to the baseline configuration.

Like the rolling moment coefficient, the yawing moment coefficient shows a significant difference between the configurations in the order of magnitude for the yawing moment coefficient at high lift conditions as seen in Figure 16a.

The yawing moment coefficient magnitude decreases for the twisted configuration and at lower lift conditions the change is even smaller. Therefore, the twisted configuration allows for the wing to takeoff without having any significant yawing moment, unlike the baseline configuration, which would experience significant yawing moment for a takeoff lift condition. Having the reduction in the yawing moment also reduces the need for control surfaces that produce drag in order to maintain yaw control. Typically, a vertical tail and rudder is used on traditional configurations, but for tail-less

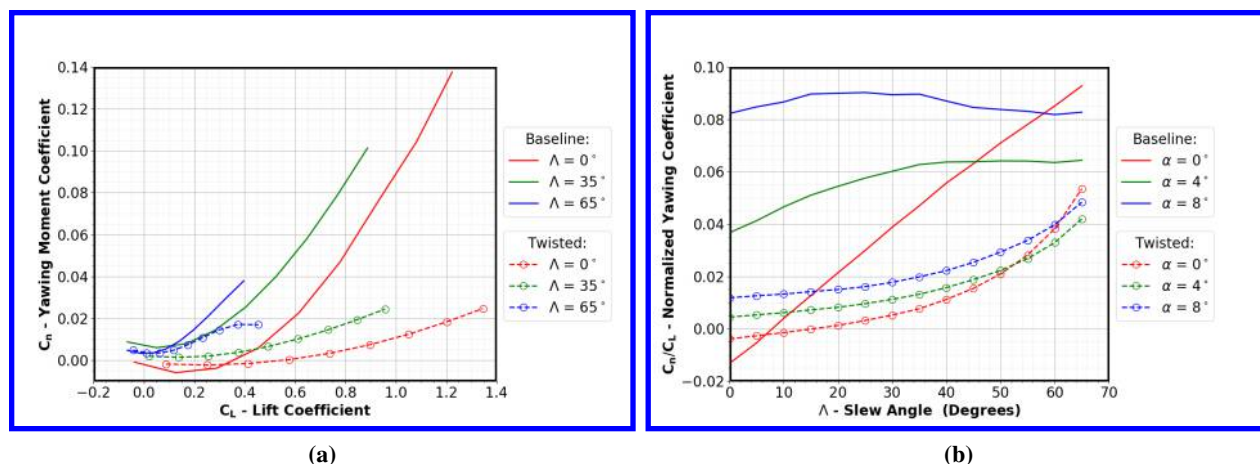


Fig. 16 a) Yawing moment coefficient comparing the baseline and twisted OFW configuration at three slew angles b) Normalized yawing moment coefficient comparing the baseline and twisted OFW configuration at three angles of attack

configurations, most of the yaw control comes from clamshell ailerons or similar control surfaces, which use drag production as a means to control the airplane. Reducing the yawing moment coefficient will reduce the amount of drag, and potentially weight, needed to control the OFW, reducing the apparent penalty in aerodynamic efficiency as shown previously.

An additional benefit that is shown in Figure 16b is the consistency in the trend of the normalized yawing moment with angle of attack for the twisted configuration. The baseline shows both significant changes in the magnitude of the normalized yawing moment and changes in the trend with respect to angle of attack. There are more similarities between the $\alpha = 4^\circ$ and $\alpha = 8^\circ$, but there is an increase in the normalized yawing moment.

D. Local Lift and Drag Coefficient Results

A greater understanding of the effects of the twist distribution can be seen in the lift distribution and drag distribution profiles for each condition. From both the Horten brothers and Prandtl, the existence of proverse yaw, and the upwash at the wingtip is derived from the lift distribution. If the lift distribution does not match the bell shaped profile, then it is expected that proverse yaw will not be efficiently generated. Both the Horten brothers and Prandtl have shown models to predict proverse yaw from a given lift distribution. However, those models have the inherent assumption of only accounting for inviscid forces, neglecting pressure and viscous forces.

Using functionality from FUN3D, the spanwise variation of the forces on both of the OFW configurations are used to determine the effects of the twist distribution on the local lift and drag forces. Initially, the $\Lambda = 0^\circ$ slew case is examined to see if the lift distributions calculated by FUN3D match those that were solved for using linear methods. The baseline and twisted lift distributions are normalized by the local chord and the total vehicle lift coefficient in order to compare differences in the lift distributions between the configurations. The baseline lift distribution is not expected to follow the bell-shaped lift distribution, but there should be some changes in the twisted distribution which make it more similar to the bell-shaped lift distribution. From before, the change in the yaw moment coefficient is caused by the geometric twist distribution, which changes both the lift and drag distribution over the OFW, but a bell-shaped lift distribution will indicate there is proverse yaw. From Figure 17a, the twisted configuration does not match the bell-shaped lift distribution. It is still evident that the asymmetries play a large role in the aerodynamic behavior of the OFW and are not easily negated with linear assumptions. For the twisted configuration, it is shown that most of the lift moves towards the center of the wing, which explains the reduction in the yawing moment coefficient. Less outboard lift production, relative to the lift production at the root of the wing, will change the drag distribution over the OFW, changing the behavior of the yaw moment coefficient.

As the slew angle is increased in Figure 17b and 17c, the maximum lift for both the baseline and twisted configuration moves backwards, especially at higher angles of attack. This behavior was also seen in the work done by R.T. Jones [1], who showed that for an oblique wing, the center of lift will move aft with increasing slew angle. An expected effect of slew angle on the OFW is that the lift distribution will change from the bell-shaped distribution as the prescribed

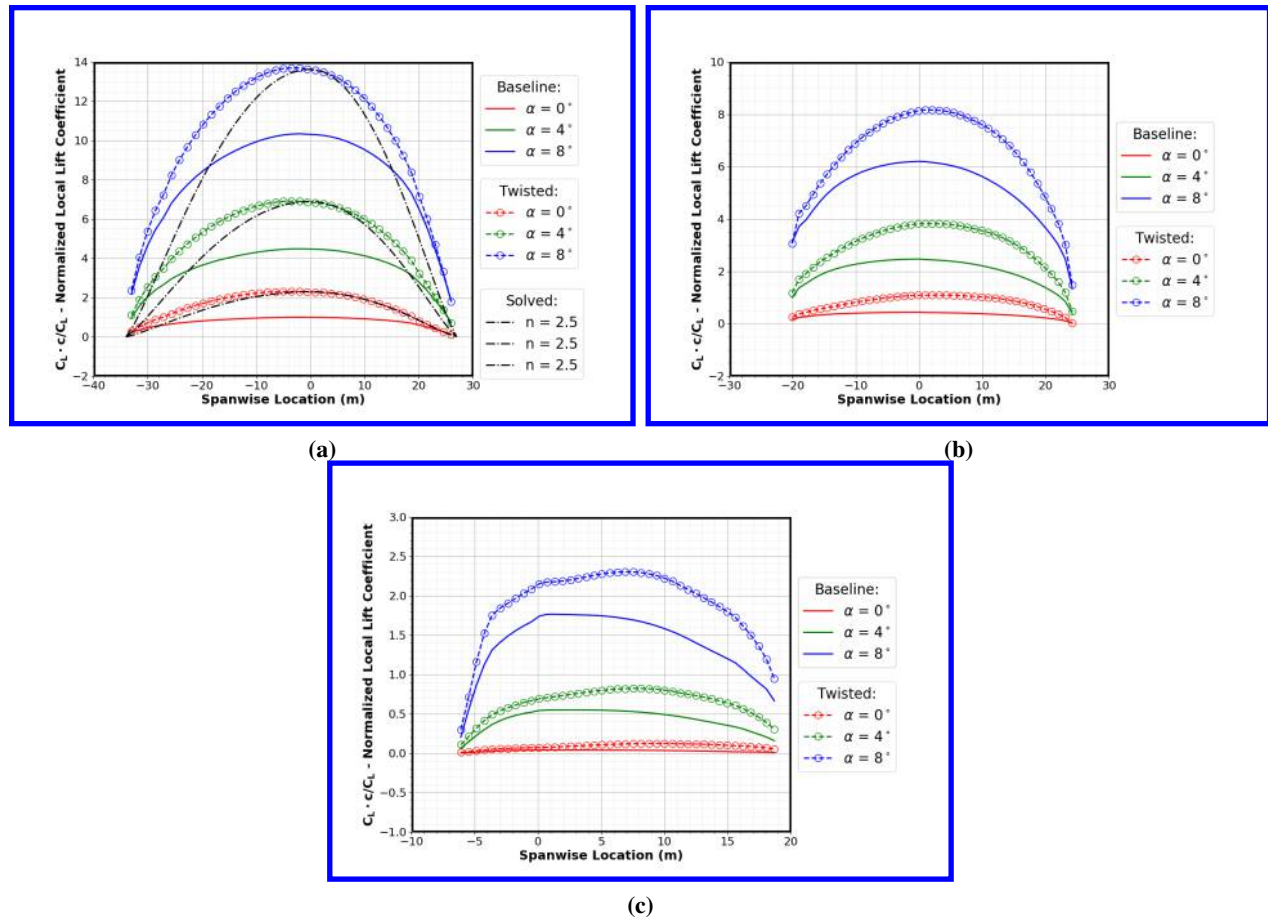


Fig. 17 Lift distributions for slewing angles of a) $\Lambda = 0^\circ$ b) $\Lambda = 35^\circ$ c) $\Lambda = 65^\circ$

twist distribution is determined for one slewing angle, in this case, the slewing angle is $\Lambda = 0^\circ$. Like the force and moment coefficient results, the slewing of the OFW reduces the differences between the baseline and twist configuration. At lower angles of attack for the $\Lambda = 65^\circ$ case, the lift distributions are nearly identical, and at higher angles of attack, the profiles are more similar in shape with only some changes in the magnitude.

A final check for the existence of proverse yaw is done by examining the drag distribution profiles to see if there is in fact negative drag at the wingtips, which is unlikely given the lift distributions seen previously. Like the lift distributions, the local drag distributions are normalized by the total drag of the OFW and only positive angles of attack considered to ensure there is a positive lift coefficient. Figure 18a shows the drag distributions for the $\Lambda = 0^\circ$ cases of the baseline and twisted configurations. Like the lift distributions, there are similarities in the profile shapes, with the twisted configuration showing greater drag at the center of the OFW, which is also seen in the lift distributions.

The primary difference in the drag distributions comes in the inboard sections which show that the drag is greater for the twisted configuration. For the $\Lambda = 0^\circ$ case, on the left side of the OFW, there is negative drag for both configurations, showing that there is some proverse yaw at the wingtip, but the proverse yaw is also seen in the baseline configuration. On the right side of the OFW, there is also a reduction in the drag, but does not become proverse. As the OFW is slewed, the drag on the left portion of the wing is dramatically reduced compared to the center and right side of the OFW. At a slewing angle of $\Lambda = 35^\circ$ and an angle of attack of $\alpha = 0^\circ$, there is proverse yaw at the left outboard section of the wing, which is not seen in the other angle of attack cases.

As the slewing angle is increased, the majority of the drag produced shifts to the right side of the wing. This would cause an increase in the yaw right tendency of the OFW, especially when the drag on the right and left side of the OFW are compared. With increasing slewing angle, the magnitude of the drag on the right side also increases, which will also have a significant impact on the yaw moment. This trend is easily captured in Figure 16a as the magnitude of the yaw moment increases with slewing angle for a given lift coefficient for both configurations. Additionally, the drag profiles

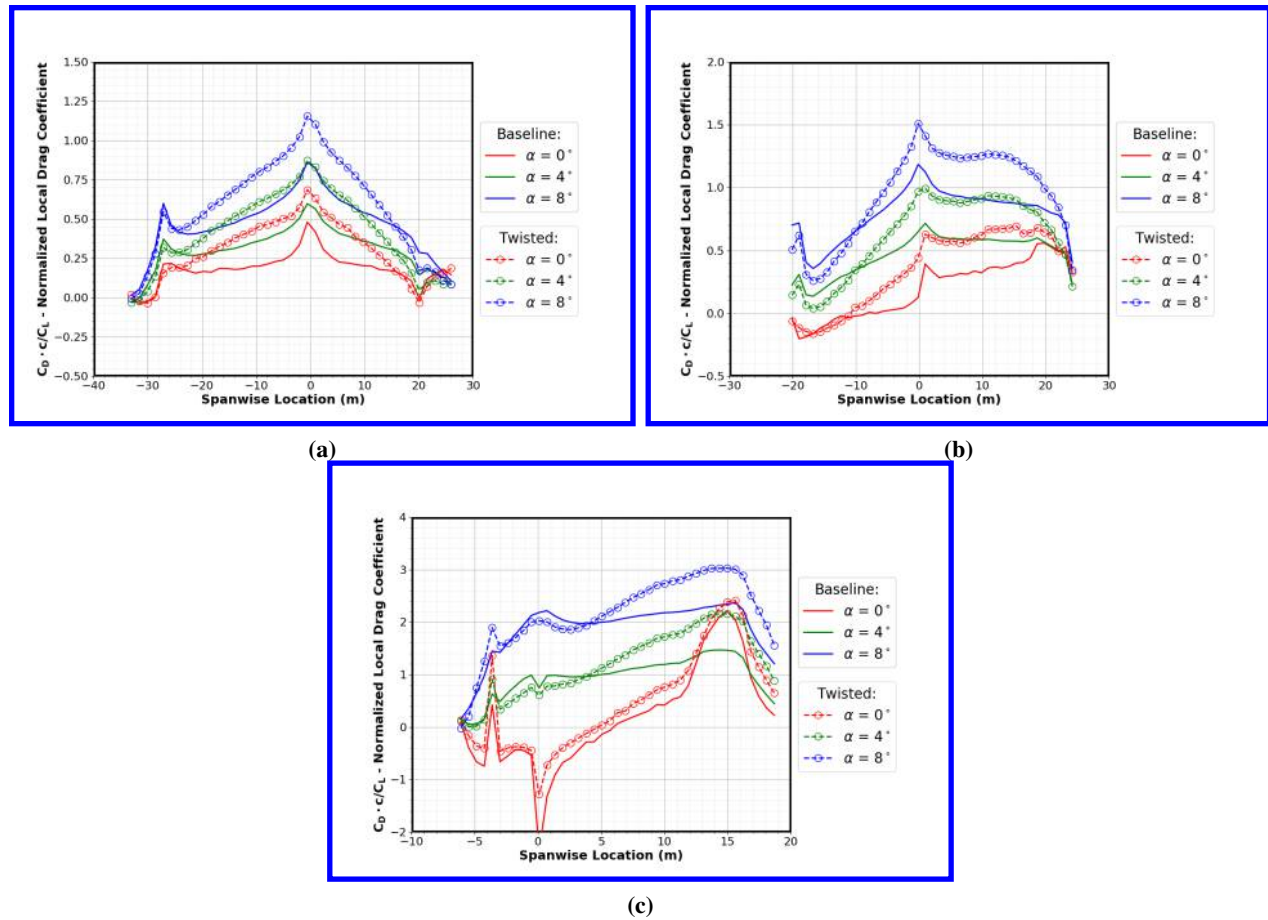


Fig. 18 Drag distributions for slew angles of a) $\Lambda = 0^\circ$ b) $\Lambda = 35^\circ$ c) $\Lambda = 65^\circ$

become more similar across the span of the OFW with increasing slew angle, which is seen throughout the force and moment coefficients. From the local drag distributions, it is clear that the bell-shaped lift distribution was not created as the lift distribution does not follow the bell-shaped profile and the drag profile does not show significant proverse yaw at either wingtip. However, majority of the changes due to twist distribution was created inboard which did have a significant impact on the behavior of the OFW.

VI. Conclusions

The primary objective of this research was to understand the impact of slew angle and twist distribution on the forces and moments for an OFW configuration. Implementing a twist distribution allowed for many of the unfavorable aerodynamic moments to be negated, allowing for a more controllable configuration, especially in a low Mach number flight envelope. While a secondary goal of implementing a twist distribution that produces a bell-shaped lift distribution fell short of conclusive, the impact of the prescribed twist distribution on the total vehicle performance cannot be ignored. It is clear that with the complex planform and asymmetry, more physics needs to be accounted for when attempting to prescribe a lift distribution with a fixed planform; crude linear assumptions are not enough, but gave great insight into the behavior of the OFW. While proverse yaw was not achieved, the twisted OFW configuration shows many of the benefits that come with a tailored lift distribution:

- The twisted OFW configuration shows a reduction in the magnitude of the rolling and yaw moment coefficients and greater consistency in the trends with both angle of attack and slew angle, reducing the impact of the asymmetries on the stability of the OFW and possibly control.
- There is an increase in the pitching moment magnitude for the twisted configuration and becomes unstable at slew angles greater than $\Lambda = 60^\circ$. Considering the Mach number flight regime and taking into account the need

for trimmed level flight, it is unlikely that the OFW configuration would fly at a high slew angle and low Mach number.

- Untrimmed aerodynamic performance is reduced for the twisted OFW configuration at $\Lambda = 0^\circ$ by 27.9% at $\alpha = 0^\circ$ and by 14.8% at $\alpha = 8^\circ$, which is to the detriment of the configuration, but directional stability is gained, negating the loss in efficiency with increased stability.
- Both configurations show that at $\Lambda = 35^\circ$ proverse yaw is generated at the left wingtip, and at $\Lambda = 0^\circ$, some proverse yaw is generated at the left wingtip. This indicates the potential of a twist distribution that could be tailored for the bell-shaped profile.
- Achieving the bell shaped lift distribution with geometric twist and taper at a single slew angle does not result in similar distributions at other slew angles. Incorporating a compliant leading and/or trailing edge control surface which can actively change the camber of the effective airfoils could allow tailoring of the lift distribution for multiple slew angles.

Overall, there is much to gain in terms of control of the moment coefficients when a twist distribution is implemented into the OFW configuration. It is yet to be determined how impactful the improvements in the rolling and yawing moments are considering the issues that still linger with the pitching moment, but opening the design space for the configuration could lead to a novel aerodynamic and/or geometric twist distribution that is able to improve on the pitching moment.

VII. Future Work

Future work will be done to validate the force and moment results with low-speed wind tunnel testing on the configurations of interest. A revisit is required to determine a twist distribution method which can be used to efficiently and consistently generate the bell-shaped lift distribution. Exploring the aerodynamic twist, along with an investigation into the impact of different airfoils on the moment coefficients, might yield more favorable results which will be included in the revisit for determining a bell-shaped lift distribution. Additionally, the transonic and supersonic performance of the bell-shaped lift distribution has yet to be explored for the OFW configuration, which will be conducted using higher-fidelity CFD to understand the trade-off in aerodynamic performance and stability. Ultimately, the addition of flexible leading and trailing edge morphing surfaces would help tailor the distribution of aerodynamic forces for the most efficient means of achieving vehicle stability and control, while minimizing any aerodynamic efficiency penalties associated with a fixed twist distribution.

VIII. Acknowledgments

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